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PALEOMAGNETIC EVIDENCE ON THE ORIGIN OF THE MARQUETTE AND STEEP ROCK HARD HEMATITE AND GOETHITE DEPOSITS

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Hard hematite ore is mined at the Cliff-Shaft mine, Marquette Range, Michigan, and hard goethite ore is mined at the Caland and Steep Rock mines, Steep Rock Range, Ontario. A paleomagnetic study was made on 67 samples from the ore horizons and associated rocks of these deposits. Alternating-current demagnetization and other standard paleomagnetic procedures were used. Geologists have proposed several theories to explain the genesis of these iron deposits of direct-shipping grade. Both ore deposits have a prefolding remanence that differs in age from any of the associated rock units. This indicates that both deposits were formed as laterites resulting from the oxidation and leaching of primary iron formation by meteoric solutions. The hard hematite ore was derived from the primary magnetite-banded oxide facies of iron formation which underlies the ore at the Cliff-Shaft mine, and the hard goethite ore was derived from the primary pyrite facies at the Caland or Steep Rock mines. The laterites were subsequently buried, folded, faulted, and intruded, but the mineralogy, structure, and texture of the ores were essentially unchanged during these events.

INTRODUCTION

Paleomagnetism can be used to determine the genesis of some iron ore deposits by comparing the remanence direction of the ore deposit with that of the host rock or of other rocks of known age from the same geologic province, and by determining whether the remanence of the ore deposit was acquired before or after folding. Thus a syngenetic deposit has a prefolding remanence which it acquired during the time interval between deposition of the enclosing rock formations. By this method, Gross and Strangway (1961) concluded that the "deep hard hematite" deposit at F'Derik in Mauritania is a Precambrian syngenetic deposit and that the "surficial hard hematite" deposit at Rouessa West is a Cenozoic laterite. Similarly, DuBois (1961, 1962) demonstrated that a hard hematite deposit in Cumberland, England, is an epigenetic deposit of Permian–Triassic age in host rocks of Lower Carboniferous age. In this study, paleomagnetism is used to study the genesis of the hard hematite ore of the Cliff-Shaft mine, Marquette Iron Range, Michigan, U.S.A., and of the hard goethite ore of the Caland and Steep Rock mines, Steep Rock Iron Range, Ontario, Canada.

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PALEOMAGNETIC METHODS

The two major reference texts in paleomagnetism are those of Irving (1964) and Nagata (1961). Detailed discussions of the equipment and procedures used in this study are given in Symons (1965).

A large, oriented hand sample was collected at each sampling site. The samples are numbered in the order in which they were collected at each locality, and the locality is indicated by the prefix letter, which is A, R, or S for the Cliff-Shaft, Caland, or Steep Rock mines, respectively. The samples were oriented by eye in relation to surveyed features on the mine maps such as drifts, roadways, etc., and the orientation is estimated to be accurate to within 5° of arc at the 95% confidence level. From each hand sample, a sample of suitable size was broken off by a hydraulic press, oriented and encased in the center of a 4-in. cube of plaster of paris. The magnetic measurements were made on the encased samples by an astatic magnetometer with a sensitivity of 5×10^{-8} e.m.u./g and a system of measurement similar to that described by As (1960). The alternating field (a-c.) demagnetization equipment had a maximum field intensity of 1 000 oersteds (peak), three sets of Helmholtz coils to null the earth's magnetic field, and utilized a variac and a sample withdrawal to decrease the alternating field. The samples were demagnetized along three perpendicular axes. All the calculations were processed on an IBM 7094 computer. The statistics follow the equations of Fisher (1953).

First, the natural remanent magnetization (n.r.m.) of each sample was measured. Next, seven samples were subjected to a-c. demagnetization in steps (Fig. 1). The remaining samples were a-c. demagnetized at a field intensity of 700 oersteds (peak), because the intensity and direction of remanence of these seven samples changed only slightly between the 500 and 700 oersted steps. The seven samples were also examined in polished sections. Eleven samples with the least stable and least intense remanence were measured after a-c. demagnetization at 720 oersteds, with the sample's orientation reversed for every procedural operation during demagnetizing and remeasuring. An unchanged remanence between the 700 and 720 oersted steps indicates that the remanence is stable.

The change in the orientation of the remanence directions for rocks of known age in North America is illustrated as a path in Fig. 2 for the period of time from the Lower Keweenawan to the present. This path was taken from the compilations of all published results given by DuBois (1962), Irving (1964), and Strangway (1964). Strangway (1964), Fahrig *et al.* (1965), and Symons (1965) cited evidence for the continued change in remanence orientation during the Middle and Early Precambrian.

CLIFF-SHAFT MINE, MARQUETTE RANGE

Geology and Paleomagnetic Results

The geology of the Marquette Range was described by Van Hise and Bayley (1897), Van Hise and Leith (1911), Boyum (1954), and James (1958). The stratigraphic sequence given in Table I has been fitted into the subdivision of

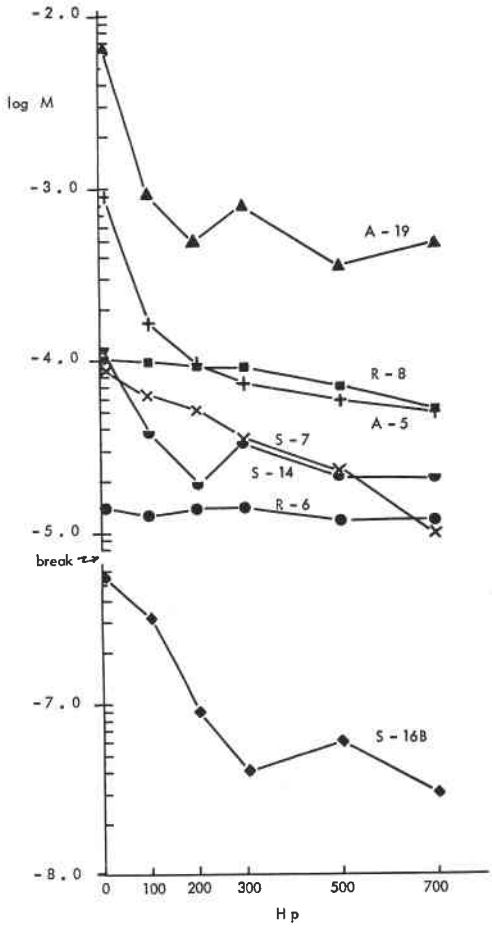


FIG. 1. Alternating-current demagnetization curves. The symbol M is the remanence intensity in e.m.u./g and H_p is the peak intensity of the a-c. demagnetizing field in oersteds. The following mineralogic descriptions are from the examination of the samples in polished sections. A-5 is from the hard hematite ore of the Cliff-Shaft mine. The very fine-grained matrix of hematite contains scattered coarse grains of euhedral magnetite which have partially altered to hematite. A-19 is from the magnetite-banded oxide facies of the Negaunee iron formation, Cliff-Shaft mine. A few fine-grained laths of hematite, which total about 1% of the polished section, are randomly oriented in the gangue. Probably the hematite has been altered from magnetite. R-6 is from the highly altered lowermost hanging wall ash rock of the Caland mine. The polished section was lost because of its softness. The hand sample has soft red hematite in spots and laminations. R-8 is from the ore zone goethite member of the Caland mine. The goethite is massive and anhedral with very minor amounts of gangue, hematite, and pyrite as inclusions. S-7 is from the unaltered hanging wall ash rock of the Steep Rock mine. Fine-grained euhedral crystals of pyrite and magnetite, which each total 1% of the section, are disseminated through the gangue. There are a few coarse-grained crystals of magnetite which are mostly altered to hematite. S-14 is from the slightly altered portion of the lower hanging wall ash rock of the Steep Rock mine. Throughout the gangue are disseminated a few euhedral laths of magnetite which show a minor amount of hematite alteration. S-16B is from a granite pebble in the basal conglomerate of the Steep Rock mine. A few crystals of pyrite are disseminated through and corroded by the gangue. There is a trace amount of magnetite and secondary chalcopyrite.

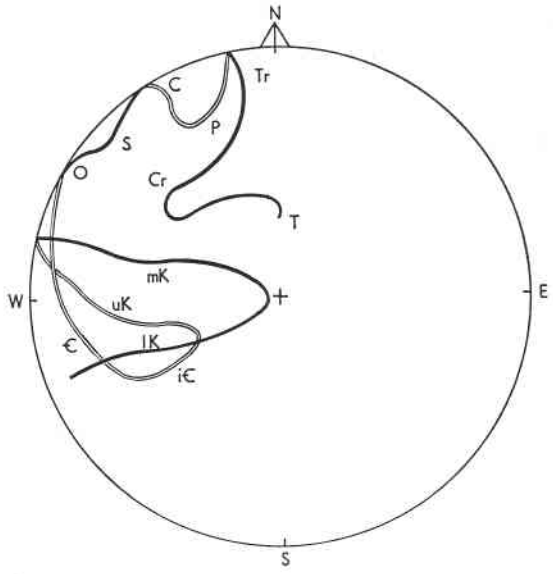


FIG. 2. Path of the mean remanence directions for rocks of the Lake Superior region. The path is plotted on a Schmidt stereographic projection with the solid (double) line joining downwardly (upwardly) directed mean remanence directions. The symbols are: T, Tertiary; Cr, Cretaceous; Tr, Triassic; P, Permian, C; Carboniferous; S, Silurian; O, Ordovician; €, Cambrian; i€, Infra-Cambrain; uK, Upper Keweenawan; mK, Middle Keweenawan; and lK, Lower Keweenawan. The data are derived from DuBois (1962), Irving (1964), and Strangway (1964).

TABLE I

Stratigraphic sequence for the Cliff-Shaft mine, Marquette Range

Era	Age (b.y.)*	Period–system (orogeny)	Marquette Range
Paleozoic	0.6	Cambrian	Sandstone
	1.0	(Grenville)	Angular unconformity
Late Precambrian		Keweenawan	Intrusive or extrusive rocks of diorite composition
	1.7	(Penokean)	Intrusive and conformable contacts
Middle Precambrian		Animikie Upper	Michigamme clastic and pyroclastic sediments
			Goodrich quartzite: basal formation of series
			Angular unconformity
		Middle	Negaunee iron formation
			Hard hematite ore: hematite-banded oxide facies(?)
			Jasper: magnetite-banded oxide facies
			Siderite and chert: nongranular silicate facies
			: carbonate facies
			Siamo slate
			Ajibik quartzite
			Angular unconformity
	2.5	Lower	Clastic sediments
			Unconformity
Early Precambrian			Granite gneiss and greenstones

*b.y., billion (10^9) years.

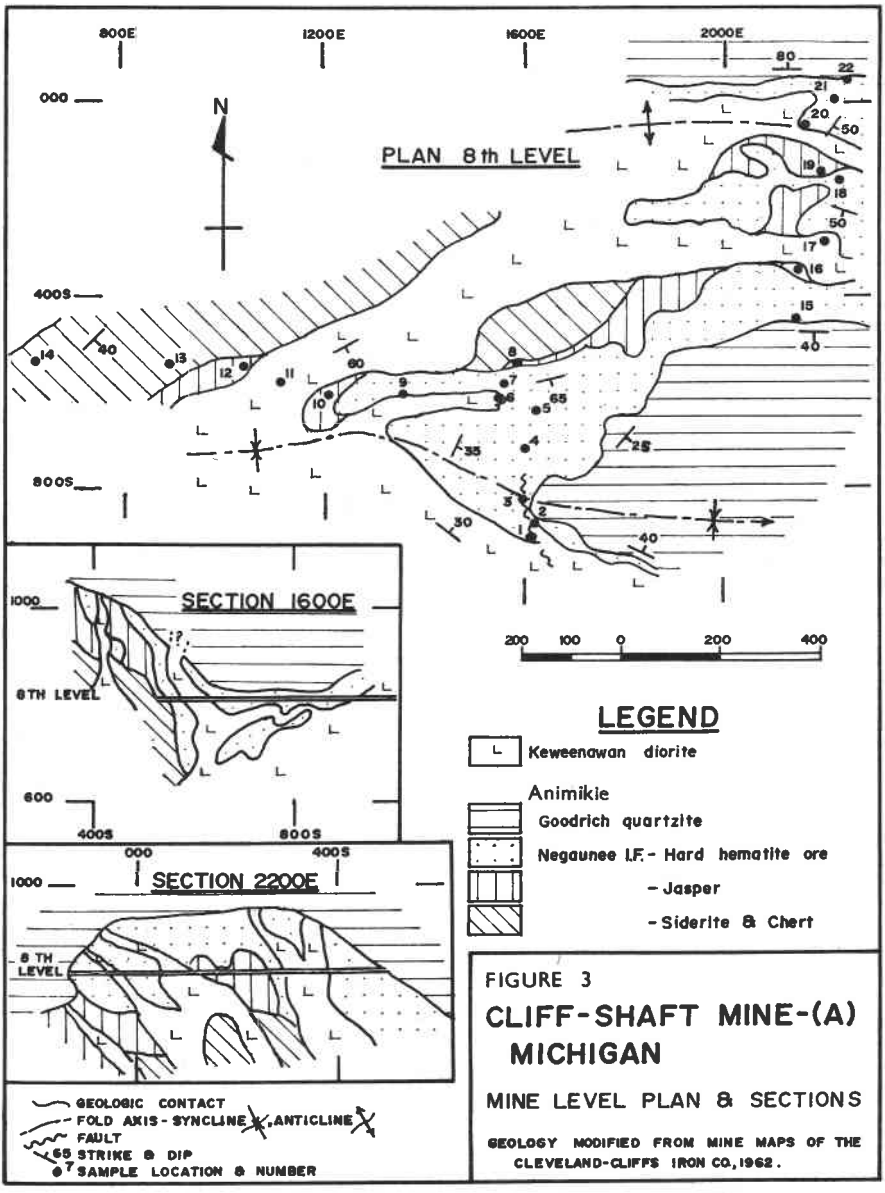


FIG. 3. Map of the Cliff-Shaft mine, Marquette Range, Michigan.

Precambrian time proposed by Goldich *et al.* (1961) for the Lake Superior region. The 27 samples from the Cliff-Shaft mine were collected underground from across a syncline–anticline structure plunging at about 25° to the east, as shown in Fig. 3, and which is part of a regional westerly pitching synclinorium. A summary of the magnetic measurements of the samples is given in Table II.

TABLE II

Summary of magnetic measurements

Sample group	N	X	Ms	Mn/Ms	Mean remanence vector					Pole position			
					R	Dm	Im	α_{63}	k	Dp	Ip	dp	dm
Cliff-Shaft mine*													
Negaunee iron formation	6(C)	3–4 900	0.05–46	1–16	4.7	169	19	20	3.8	75	−33	11	21
Hard hematite ore member	8(C)	82–850	4.3–13	3–18	7.3	239	30	10	10.3	144	−8	6	11
Keweenawan diorite	4(U)	3–95	0.30–3.2	1–6	3.3	213	58	23	4.3	113	0	25	34
Caland and Steep Rock mines†													
Laurentian granodiorite	3	0.13–0.38	0.001–0.012	7–10									
Basal conglomerate: all	6	0.006–1.1	0.003–0.010	2–20									
: matrix	4(C)				3.7	232	13	13	11.2				
Ore zone chert member	1(C)	0.06	0.05	2		301	20						
Ore zone goethite member	15(C)	1.3–2.6	0.015–3.0	1–2.5	12.4	261	53	10	5.5	152	19	10	14
South limb	4(C)				3.9	322	40	8.7	26.6				
Center limb	4(C)				3.9	254	37	7.2	38.0				
North limb	4(C)				3.7	224	56	13	11.6				
Steep Rock limb	3(C)				2.5	223	53	23	4.2				
Hanging wall ash rock	9(C)	0.40–184	0.027–2.0	1–20	8.2	243	14	9.3	9.9	156	−12	5	9
Upper sediments	1(C)	0.58	0.031	2		236	3						
Basic extrusives	5(C)	1.5–2.0	0.016–0.09	1–2	3.3	341	35	32	2.3	235	57	21	36
Basic "ore-cutting" dike	1	2.3	0.021	4		208	6						

*Longitude, 87.6° W.; latitude, 46.5° W.
†Longitude, 91.6° W.; latitude, 48.8° W.

NOTE: The symbols and units are *N*, the number of samples represented in the group or the population of the directions; C(U), indicates that the remanence directions have been corrected (uncorrected) for tectonic deformation before calculation of the mean remanence direction and pole position; *X*, the range of susceptibilities in e.m.u./g × 10^{-5}; *Ms*, the range of intensities of the stable remanences in e.m.u./g × 10^{-5} after a-c. demagnetization at 700 oersteds (peak); *Mn/Ms*, the range of ratios of n.r.m. intensity to *Ms* intensity, which roughly indicates the ratios of unstable (*Mn–Ms*) to stable (*Ms*) remanence; *R*, the length of the mean remanence vector with each of the *N* remanence directions considered as a unit vector; *Dm*, the declination of the mean remanence vector in degrees reckoned clockwise from 0°; *Im*, the inclination of the mean remanence vector in degrees reckoned from the horizontal plane and considered positive (negative) if directed downwards (upwards); α_{63} the semiangle of the circular standard deviation, in degrees, of the cluster of remanence directions; *k*, the precision parameter of the cluster of remanence directions; *Dp*(*Ip*), the longitude (latitude) of the virtual paleomagnetic pole position in degrees: *dp*(*dm*), the semiaxis of the oval of confidence derived from the circular standard deviation perpendicular to (along) the meridian between the sampling site and the pole position in degrees about the pole position.

At this mine, the Negaunee iron formation is about 2 200 ft thick; it has been metamorphosed regionally to the biotite facies (James 1955); and its minimum age from two K–Ar radioactive age dates is 1.61 or 1.89 billion (10^9) years (Goldich *et al.* 1961). Of the eight samples collected from the Negaunee iron formation, excluding the hard hematite ore member, one (A-8) gave an anomalous remanence direction and one (A-16) had unstable remanence. The remaining six samples represent the magnetite-banded oxide facies or jasper (A-10, 12, and 19), the carbonate facies (A-13 and 14), and the nongranular silicate facies (A-3) as defined by James (1954). These six samples have a wide range of magnetic properties (Table II), which is expected, considering the variety of facies represented, and which suggests that the remanence is related to the primary sediment. James (1954) considered that the different facies of iron formation simply represent changes in the chemical environment within the marine depositional basin, and that they are all chemical sediments. Lepp and Goldich (1964) proposed that chemical transformations occurred in a common "proto-iron formation" to form the different facies of iron formation. These two papers present the two major hypotheses on the genesis of iron formation, but in either hypothesis the resultant iron formation would be expected to have a chemical remanent magnetization (c.r.m.). The a-c. demagnetizing curve for one sample (A-19) of magnetite-banded oxide facies is given in Fig. 1. This sample has an unstable remanence moment as indicated by the drop in remanence intensity between the 0 and 100 oersted steps, which probably resides in the magnetite for the most part, and a stable remanence moment as indicated by the unchanged remanence intensity between the 200 and 700 oersted steps, which probably resides in the hematite for the most part. Tectonic correction or unfolding the structure to a flat plane causes the precision parameter to decrease slightly from 5.3 to 3.8. The precision parameter is a measure of the statistical significance which can be attached to a geological grouping of the remanence directions. In general terms, the higher the value of the precision parameter, the greater is the significance that can be attached to the grouping, and any value in excess of 3.0 has significance. While the postfolding remanence for the iron formation has a higher statistical significance than the prefolding remanence (that is 5.3 to 3.8), the difference is relatively small, so that the fold test is not conclusive for the iron formation. The range of magnetic properties favors the interpretation that the samples have a primary, prefolding c.r.m. in the direction shown in Fig. 7.

The hard hematite ore member of the Negaunee iron formation is less than 100 ft thick; it occurs only locally and at the very top of the iron formation; and it is composed of massive specular hematite with minor magnetite and with minor secondary soft hematite and silica (Fig. 6*a*) (Han 1962, personal communication). Two samples (A-21 and 22) of hard hematite ore proved to have unstable remanence. The remaining eight samples (A-1, 4, 5, 7, 9, 15, 18, and 20) proved to have quite uniform magnetic properties (Table II). After removal of the small unstable moments, which most probably originate in the very minor amounts of magnetite, all the samples have very stable remanences of weak intensity, which are most probably c.r.m. moments originating in the

hematite (Fig. 1, sample A-5). Unfolding causes the precision parameter to increase from 5.6 to 10.3 when all eight samples form the group, from 10.7 to 20.7 when the five samples (A-1, 4, 5, 7, and 9) from across the synclinal fold (Fig. 3) form the group, and from 1.8 to 43.5 when the two samples (A-18 and 20) on either side of the anticlinal fold form the group. The marked increase in the precision parameter in each group clearly indicates that the hard hematite ore acquired its remanence before folding and in the direction shown in Fig. 7.

The Michigamme series overlies the Negaunee iron formation with an angular unconformity of less than 15°; it comprises over 3 000 ft of clastic and pyroclastic sediments; and it forms the ore's hanging wall.

The diorite bodies associated with the Negaunee iron formation have been interpreted to be intrusives of Keweenawan, Penokean, or Upper Animikian age, or extrusives of Middle Animikian age. They occur as sills up to several hundred feet thick and cross-cutting dikes. As shown in Fig. 3, the four diorite samples (A-2, 6, 11, and 17) come from portions of the diorite which are essentially conformable with the bedding of the iron formation. The four samples have the very stable remanences and the other magnetic characteristics (Table II) which would be expected of a thermal remanent magnetization (t.r.m.). Unfolding causes the precision parameter to decrease from a statistically significant 4.3 to an insignificant 1.3, which strongly indicates that the diorite acquired its remanence after the folding of the Animikie strata. Also, the mean remanence direction before tectonic correction (Fig. 7) is very similar to other directions reported by DuBois (1962), Strangway (1964), Fahrig *et al.* (1965), and Symons (1965) for Lower to Middle Keweenawan rocks (Fig. 2).

Discussion of Results

James (1954) proposed that the hard hematite ore was deposited as a chemical sediment of the hematite-banded oxide facies of iron formation under intense oxidizing conditions at the end of the Middle Animikie. Royce (1932) argued that meteoric solutions oxidized the siderite and leached the chert of the primary iron formation after the Middle Animikie to leave residual soft hematite deposits on the surface, which were subsequently buried and metamorphosed to hard hematite. Van Hise and Leith (with Mead) (1911) suggested that, after burial, meteoric solutions of Upper Animikian age oxidized and leached the iron formation to leave soft hematite deposits in structural troughs. Later metamorphism and intrusive activity converted the soft hematite to hard hematite. Roberts and Bartley (1943*a*) proposed that iron-bearing hydrothermal solutions ascended from a magmatic source at the end of the Middle Animikie and deposited the iron by replacement of the primary iron formation to form a surficial, soft hematite deposit. This deposit was converted to a hard hematite laterite, which was subsequently anamorphosed. Gruner (1930, 1937), Adler (1935), and Boyum (1954) supported postfolding hydrothermal hypotheses for the genesis of the hard hematite ore.

The paleomagnetic results place limitations on any hypothesis of ore genesis. The Negaunee iron formation must have acquired its prefolding remanence,

probably a c.r.m., during either sedimentation or diagenesis in the Middle Animikie. The hard hematite ore also has a prefolding remanence, probably a c.r.m., which deviates in direction by 70° of arc from that of the underlying conformable iron formation. This deviation represents a time interval of tens to hundreds of millions of years between deposition of the iron formation and formation of the ore, which contradicts James' (1954) hypothesis of ore genesis. Further, the form and mineralogy of the hard hematite ore cannot have been altered appreciably since formation, otherwise its prefolding nature would not have been preserved. Its preservation is difficult to reconcile with the hypotheses of Royce (1932) and of Van Hise and Leith (1911). Gruner (1946) noted that the oxidation and leaching of magnetite-banded iron formation will produce hematite. Thus, it seems most likely that meteoric solutions oxidized and leached surficial exposures of the magnetite-banded oxide facies of the Negaunee iron formation under intense oxidizing conditions, and converted them directly into hard hematite laterites. Gross and Strangway (1961) reported on a modern example of such a lateritic hard hematite deposit at Rouessa West in Mauritania which has yet to be reburied. The hypothesis of Roberts and Bartley (1943*a*) fulfills the paleomagnetic requirements. It utilizes hydrothermal instead of meteoric solutions. The lack of modern examples of such deposits being formed and of other evidence of volcanic activity between the Middle and Upper Animikie militates against their hypothesis. Also, a hydrothermal replacement deposit would be expected to show replacement in several stratigraphic horizons, replacement of primary textures and structures such as bedding, and a strong relationship to a vein structure, features which Smith (1919) and DuBois (1962) described in the Cumberland hard hematite deposits. In contrast, the Cliff-Shaft deposit is restricted to the uppermost horizon only, shows no evidence of replacement of textures or structures, and shows no relation to a vein structure. After formation, the hard hematite laterite was buried with a slight angular unconformity by the Upper Animikian Michigamme series. The Animikie strata were folded during the Penokean orogeny. Then the diorites were intruded into the folded strata as sills and dikes, during the Lower or Middle Keweenawan. This sequence of events accounts for the diorites' postfolding remanence, likely a t.r.m., which corresponds in direction to other Lower and Middle Keweenawan directions while differing from those of the hard hematite ore and iron formation by 33° and 55°, respectively. These dynamic and thermal metamorphic events have not altered appreciably the remanence of the hard hematite ore or iron formation. This proposed sequence of events contains elements proposed mainly by Royce (1932) and Roberts and Bartley (1943*a*), and it is consistent with the observed geologic and paleomagnetic evidence.

CALAND AND STEEP ROCK MINES, STEEP ROCK RANGE

Geology and Paleomagnetic Results

The geology of the Steep Rock Range was described and discussed by Moore (1940), Bartley (1940, 1948), Tanton (1941, 1946), Roberts and Bartley

TABLE III

Stratigraphic sequence for the Caland and Steep Rock mines, Steep Rock Range

Era	Age (b.y.)*	Period–system (orogeny)	Steep Rock Range
	2.5	(Algoman)	Granitic intrusives and granite gneiss complex
			Intrusive contact
Early Precambrian			Steep Rock group (?)
			Basic extrusives (and intrusives?)
			Paraconformity?
			Steep Rock group
			Upper sediments (clastic and volcanic)
			Hanging wall ash rock
			Ore zone: goethite member
			: chert member
			Footwall paint rock
			Disconformity
			Carbonate sediments
			Basal conglomerate
			Nonconformity
		(Laurentian)	Basement complex
			Basic dikes
			Granodioritic gneisses

*b.y., billion (10^9) years.

(1943*b*), Jolliffe (1955), Spat (1961), Gauvin (1961), and Wright (1965). The geology given here and the stratigraphic sequence given in Table III was taken mostly from the latter four reports. All 26 samples from the Caland mine and 17 samples from the Steep Rock mine were collected in the bottom of the mine open pits, except for three samples from a railway cut (S-15 to 17), two samples from road cuts (S-10 and 13), and two samples from surface outcrops (S-11 and 12). The samples from the Caland mine were collected from around the fold structure as shown in Fig. 4. This fold structure has three limbs: the south limb, which strikes N. 20° W. and dips steeply to the west; the center limb, which strikes E.–W. and dips steeply to the south; and the north limb, which strikes N.–S. and dips steeply to the west. The Steep Rock ore deposit or Steep Rock limb is a faulted extension, lying 1 mi to the west, of the Caland ore deposit as shown in Fig. 5. The Steep Rock limb strikes N.–S. with a gentle flexure and dips steeply to the west, which is very similar to the attitude of the north limb of the Caland mine. The tops of the beds of the Steep Rock group in Figs. 4 and 5 face to the southwest, as determined from grain gradations, pillow structures, and algal structures, except for one outcrop (Fig. 5, sample S-12) which shows reversed sedimentary grain gradations. Most of the faults strike northeasterly, dip steeply, and have dominantly lateral left-handed displacements with strike separations of up to 2 mi. Lepp and Goldich (1964) reported one K–Ar radioactive age date of 2.7 billion (10^9) years from the Steep Rock group, which gives its minimum age and places it definitely in the Early Precambrian. The magnetic measurements made on the samples are summarized in Table II.

All three samples (R-1 and 26; S-15) from the Laurentian granodiorite, which forms the basement complex for the region, are excluded because they had remanence intensities too weak to be measured.

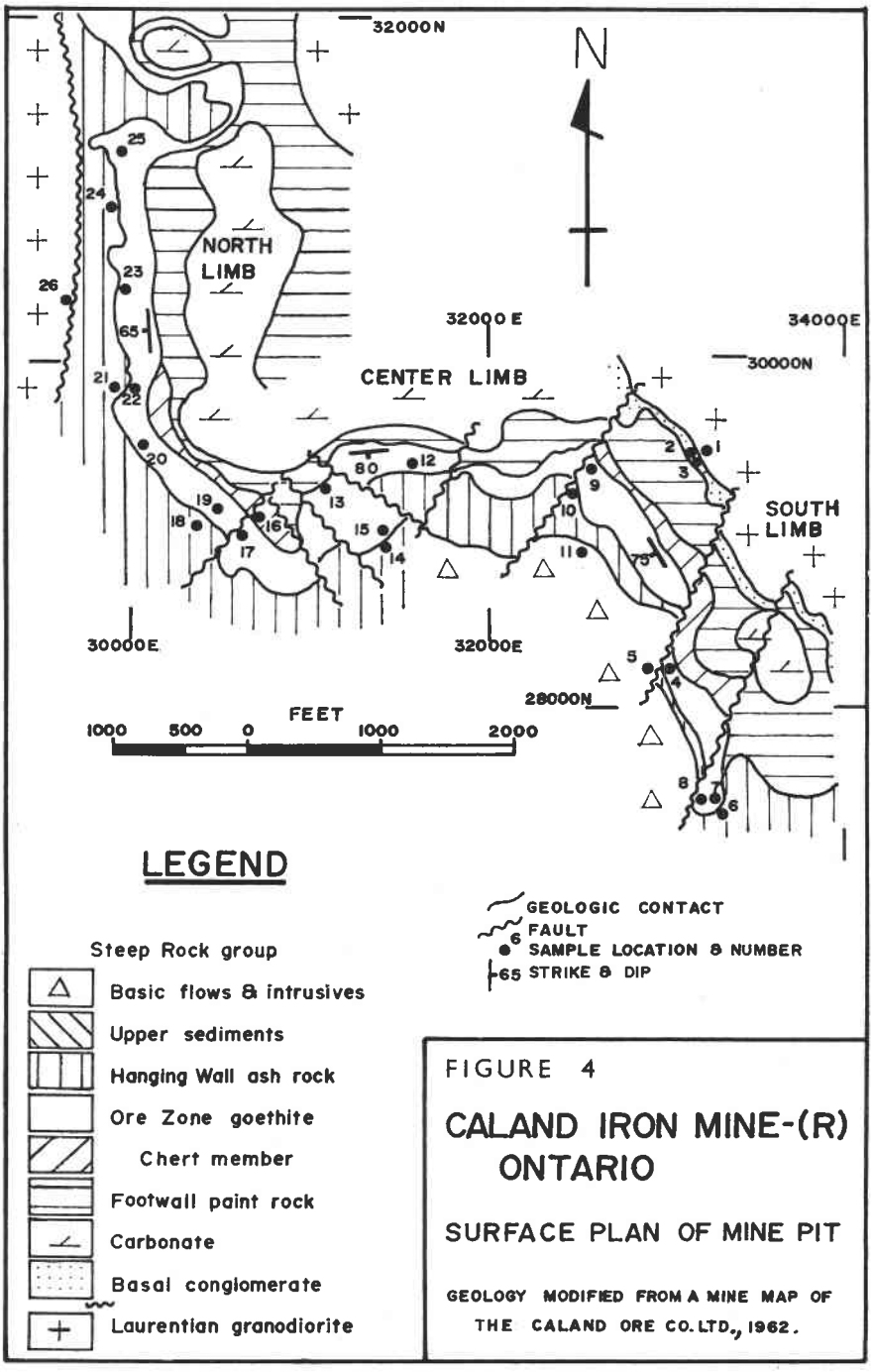


FIG. 4. Map of the Caland mine, Steep Rock Range, Ontario.

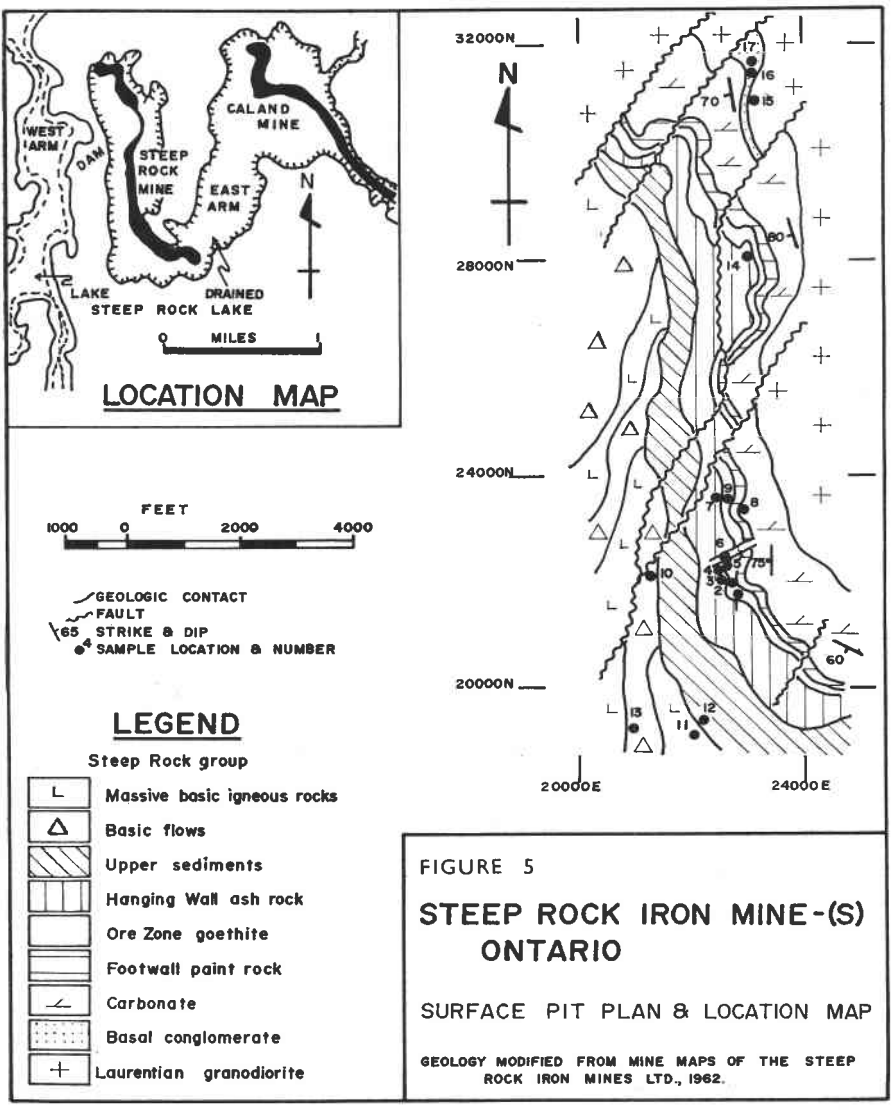


FIG. 5. Map of the Steep Rock mine, Steep Rock Range, Ontario.

The results from the ore zone goethite of the Steep Rock group are discussed first because this unit was the most thoroughly sampled. The ore zone goethite varies between 0 and 1 000 ft and averages 300 ft in thickness; it is stratigraphically continuous over a length of 7 mi and a depth in excess of 2 100 ft; it contains hard, dense goethite with colloform and pisolitic structures and with minor amounts of hematite, limonite, and quartz; and it contains pyritiferous wastebands of limonite, pisolitic aluminous clay, and thinly banded chert. One sample (R-15) gave an anomalous remanence direction. Of the remaining 15 samples, four samples (R-4, 7, 8, and 9) came from the south

PLATE I

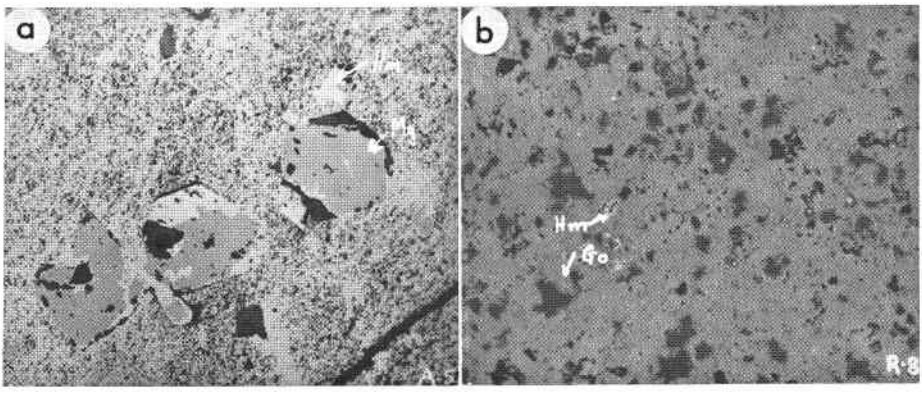


FIG. 6. Polished-section microphotographs. (*a*) This hard hematite ore sample (A-5) from the Cliff-Shaft mine is magnified by 88×. The very fine-grained hematite matrix contains scattered coarse grains of euhedral magnetite which have been partially altered to hematite. (*b*) This hard goethite ore sample (R-8) from the Caland mine is magnified by 175×. The massive anhedral goethite contains gangue and hematite inclusions. A few pyrite inclusions were noted.

limb, four samples (R-12, 13, 17, and 19) came from the center limb, four samples (R-20, 22, 23, and 25) came from the north limb, and three samples (S-1, 2, and 9) came from the Steep Rock limb. These samples have very uniform magnetic characteristics (Table II) with very stable remanences of moderately weak intensity, as shown in Fig. 1 (sample R-8), where the a-c. demagnetization has had very little effect on the remanence. The remanence is almost certainly a c.r.m. associated with the massive anhedral goethite (Fig. 6*b*), or possibly with the trace amounts of intimately associated hematite. An X-ray analysis (Hedley 1966, personal communication) showed that this sample contains mainly goethite and quartz but only a small percentage of hematite or magnetite. This result confirms the polished section analysis (Fig. 1). Unfolding causes the precision parameter for the remanence directions of the 12 ore samples from the Caland mine to remain unchanged at 6.0.

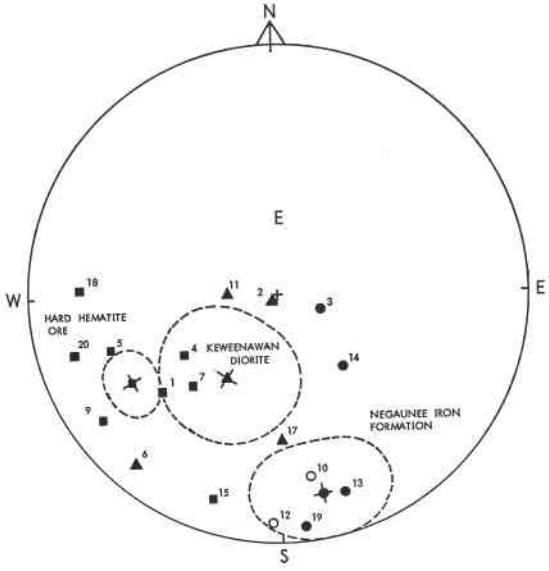


Fig. 7. Remanence directions for the samples from the Cliff-Shaft mine. The directions from the Negaunee iron formation are represented by a circle, the hard hematite ore by a square, and the Keweenawan diorite by a triangle. The directions from the iron formation and the ore are shown after tectonic correction, and those from the diorite are shown before tectonic correction. The tectonic corrections were made by rotating the bedding strike to east–west and the dip to the horizontal. The sample numbers have the prefix letter A. Conventions: The stereonet plots are Schmidt equal area polar projections viewed from above the pole. The declination of the remanence directions is reckoned clockwise from the north position. The inclination is reckoned from the primitive horizon and the direction is indicated by a solid (open) symbol if it is positive (negative) or downwardly (upwardly) directed. The direction of the present earth's magnetic field is indicated by the symbol E. The circular standard deviation is indicated by a broken line about the mean remanence direction of the group, which is indicated by a cross through the group symbol.

However, unfolding causes the precision parameter for the four ore samples from the south limb to increase from 14.0 to 26.6 (Fig. 8*a*), from the center limb to increase from 8.0 to 38.1, and from the north limb to increase from 5.5 to 11.6, which clearly indicates that the remanence is prefolding. The

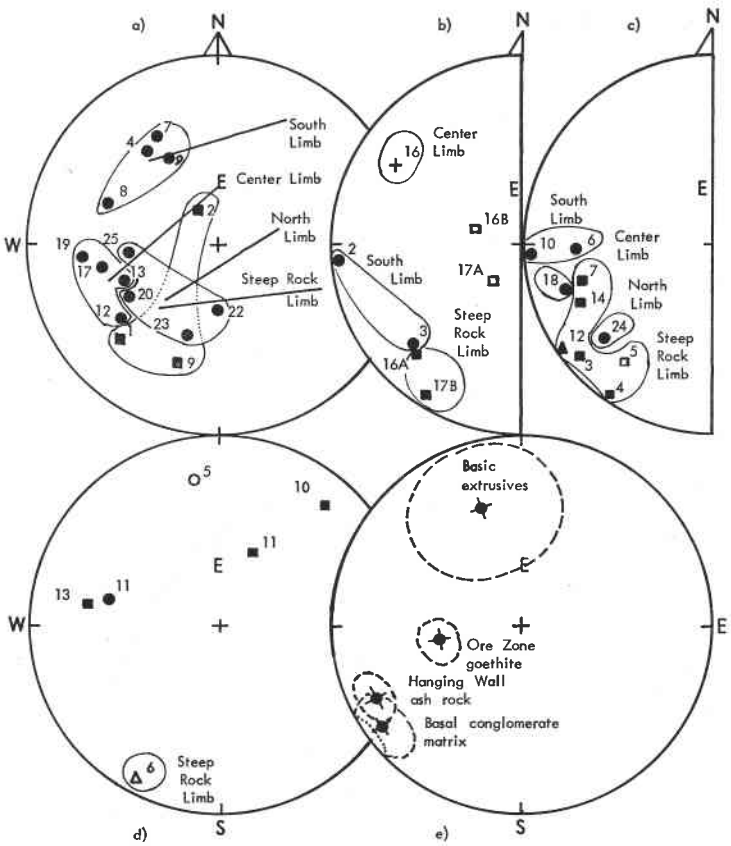


FIG. 8. Remanence directions for samples from the Steep Rock Range. All the directions are shown after tectonic correction. The corrections were made by rotating the strike to S. 20° E. and the dip to horizontal. The samples from one limb of the fold structure are enclosed by a solid line. The sample numbers from the Caland and Steep Rock mines have the prefix letters R and S, respectively. The plotting conventions are the same as for Fig. 7. (*a*) Ore zone goethite member samples from the Caland mine (circle) and Steep Rock mine (square); the straight, solid lines begin in the mean remanence direction for their respective fold limbs before tectonic correction and end in the mean remanence direction after tectonic correction, and thereby indicate the effect of applying the tectonic corrections. (*b*) Basal conglomerate samples from the Caland mine (circle) and the Steep Rock mine (square), and an ore zone chert member sample from the Caland mine (cross). (*c*) Hanging wall ash rock samples from the Caland mine (circle) and Steep Rock mine (square), and an ore-cutting altered basic dike sample from the Steep Rock mine (triangle). (*d*) Upper sediment sample from the Steep Rock mine (triangle), and basic extrusive samples from the Caland mine (circle) and Steep Rock mine (square). (*e*) Mean remanence directions of groups of samples from the Steep Rock Range.

mean remanence directions for the ore samples from each limb, after tectonic correction (Fig. 8*a*), are similar in inclination but differ significantly in declination because the limbs of the fold have been rotated. Because a tectonic correction cannot be applied to compensate for this rotation, the overall value of the precision parameter is unchanged on unfolding. If the center limb is held constant and used as a reference position, the mean remanence directions indicate that the south end of the south limb has been rotated down through

about 45° of arc around the flexure point joining the south and center limbs, and the north end of the north limb has been rotated down through about 25° of arc around the flexure point joining the north and center limbs. Unfolding causes the precision parameter for the remanence directions of the three samples from the Steep Rock limb to decrease slightly from 4.4 to 4.2, and this slight change is because these three samples have similar tectonic corrections. Note, however, that these three samples have a mean remanence direction which agrees to within 4° of arc with that obtained for the north limb. This agreement suggests that the Steep Rock limb was simply translated and not rotated with respect to the north limb, which is its logical faulted extension.

The basal conglomerate rests unconformably on the Laurentian granodiorite; it varies in thickness between 0 and 70 ft; and it grades from an angular boulder conglomerate, to a rounded pebble conglomerate, to a quartzitic grit, and finally to quartzite. Six samples (R-2 and 3; S-16A, 16B, 17A, and 17B) from the basal conglomerate were measured. All the samples had weak remanence intensities of the order of 7×10^{-8} e.m.u./g after a-c. demagnetization (Table II). However, the two samples (R-2 and 3) of the quartzitic grit member and the two samples (S-16A and 17B) of the matrix portion of the pebble conglomerate member have remanence directions which form a cluster after tectonic correction (Fig. 8*b*) with a statistically significant precision parameter of 11.2. Also, the distribution of their remanence directions shows the same effect of tectonic rotation of the south limb relative to the Steep Rock limb as does the goethite ore (Fig. 8*a*). In addition, the sample (R-3) with the least intense remanence had an unchanged remanence direction after a-c. demagnetization with the sample orientation reversed. This evidence indicates that these grit and matrix samples have a stable prefolding remanence which is probably a detrital remanent magnetization (d.r.m.). The two diorite-pebble samples (S-16B and 17A) from within the conglomerate have remanence directions which diverge by 90° of arc with those of their matrix material, which is the conglomerate test for stability, and one of these samples (S-16B) appears to have reached stability as shown by step demagnetization (Fig. 1). Thus, the two diorite-pebble samples provide additional evidence that the matrix material has a primary d.r.m.

The single sample (S-8) of carbonate sediment had too weak a remanence intensity for measurement. This unit is composed of 0 to 800 ft of massive, well-banded dolomite which has been brecciated and recrystallized in the uppermost portion.

The footwall paint rock is a weathered residuum composed of dark brown manganiferous clay (5–400 ft thick) with bands of goethite, limonite, and granular quartz sand.

One sample (R-16) from the basal chert member of the ore zone gave a stable remanence direction that is fairly similar to the directions of the nearest two samples (R-17 and 19) from the goethite member of the center limb (Fig. 8, *a* and *b*).

The samples from the hanging wall ash rock represent the three varieties

of ash rock that are present. Four samples (R-6, 10, and 24; S-3) are soft, red talcose schist or highly altered ash rock from the lowermost contact zone adjacent to the ore zone goethite, three samples (R-18; S-5 and 14) are light green chloritic fragmental or slightly altered ash rock from some tens of feet above the lowermost contact, and two samples (S-4 and 7) are dacite or unaltered basic pyroclastic. Two additional samples are excluded; one (R-14) because its remanence proved unstable, and one (R-21) because its remanence direction proved anomalous. Wright (1965) showed by geologic, polished section, and trace element studies that the pyrite zones found in the altered lower portion of the ash rock are syngenetic. Alternating-current demagnetization (Fig. 1, samples S-7 and 14) indicates that the remanence of the highly altered ash rock is highly stable and is probably a c.r.m. residing in hematite, and that of the unaltered ash rock or dacite is moderately stable and is probably a t.r.m. residing in magnetite. Despite the wide range of the magnetic characteristics of the nine accepted samples, as noted in Table II, their remanence directions all cluster in the same area of the stereonet after tectonic correction (Fig. 8*c*). Unfolding causes the precision parameter for the four samples from the Caland mine to increase slightly from 14.6 to 16.3 and that of the five samples from the Steep Rock mine to decrease slightly from 12.1 to 9.1, which is inconclusive as a test for a prefolding or postfolding remanence. However, the precision parameter after unfolding with all nine samples included in the group is a significant 9.9, which indicates the contemporaneity of their remanence formation. Also, the distribution of the remanence directions from the south, center, north, and Steep Rock limbs (Fig. 8*c*) follows the same pattern as found for the goethite ore (Fig. 8*a*), which indicates that the remanence was acquired before folding and rotation. Since these ash rock samples range from completely altered to unaltered, the finding of a predeformation remanence is strong evidence that the remanence was acquired at the time of deposition.

The upper sediments are an intercalated sequence of volcanic and clastic sediments. One sample (S-12) comes from the outcrop of upper sediments where anomalous grain gradations are found. The sample has a stable remanence direction which agrees very closely with the remanence directions for the underlying ash rock samples of the Steep Rock limb (Fig. 8*c*). This result does not support the conclusion that these beds of upper sediment are overturned, as Moore (1940) proposed.

Five samples (R-5 and 11; S-10, 11, and 13) of andesite were collected from the basic extrusives which are a thick sequence of basic pillowed lavas and massive extrusive rocks at the top of the Steep Rock group. The samples have fairly stable remanence moments of weak intensity, which are probably t.r.m., and moderate unstable moments, which were removed by a-c. demagnetization (Table II). The precision parameter both before and after unfolding is about 2.3, which is not very statistically significant. However, the remanence directions of these five samples occupy a different area of the stereonet and their mean direction differs markedly from the underlying formations (Fig. 8*d*).

One sample (S-6) comes from a highly altered basic dike which occupies a fault cutting the goethite ore in the Steep Rock limb. The sample's stable remanence direction corresponds closely to those of the ash rock of the Steep Rock limb, which suggests that the dike and ash rock are contemporaneous (Fig. 8*c*).

The mean remanence directions of the more thoroughly sampled formations are summarized in Fig. 8*e*.

Discussion of Results

Wright (1965) regarded the goethite as a syngenetic deposit precipitated in a restricted basin under anaerobic conditions, with the iron derived from the weathering of a nearby ferruginous dolomite in the Steep Rock group. Spat (1962, personal communication) thought that the deposit was syngenetic and that it could be either a primary sediment or a laterite. Jolliffe (1955) proposed that limonite was deposited on a deeply weathered topography. The limonite was itself weathered, buried, structurally tilted, and then altered to goethite by Algoman hydrothermal solutions before folding, faulting, and brecciation. Bartley (1940, 1948), Moore (1940), Tanton (1941, 1946), Smith (1942), Roberts and Bartley (1943*a*, 1943*b*), Hicks (1950), and Gruner (1956) proposed a variety of post-deformational hydrothermal and meteoric origins for the goethite ore.

Paleomagnetism indicates, and is consistent with, the following sequence of geologic events. The basal conglomerate was deposited nonconformably over the Laurentian granodiorite complex, with its matrix and sandy horizons having a stable d.r.m. in the direction indicated in Fig. 8*e*. The carbonate sediments, the primary footwall paint rock, and the primary iron formation were then deposited. The angular difference of about 50° of arc between their pole positions (Table II) indicates a time interval of tens to hundreds of millions of years between deposition of the basal conglomerate and formation of the goethite ore. The goethite ore has a stable prefolding c.r.m. Lateritic weathering of the primary ore formation under intense oxidizing conditions would account for the mineralogy, textures, and structures found in the goethite ore, paint rock, and upper carbonate sediments. The primary iron formation probably belonged to the pyrite facies, which would alter to goethite (Gruner 1956) and which explains the presence of syngenetic pyrite associated with the ore (Wright 1965). The suggestion of Moore (1940) that the primary iron formation was a ferruginous carbonate is contradicted by the fact that no siderite has been observed in or adjacent to the ore zone. The suggestion of Jolliffe (1955) that the primary iron formation was limonite is thought untenable, as no other Precambrian examples of limonitic primary iron formation are reported. The mineralogy, texture, and structure of the ore have essentially been unaltered since the laterite was formed.

The angular difference of about 30° of arc between their paleomagnetic pole positions suggests a time interval of the order of tens of millions of years between the genesis of the goethite ore and deposition of the ash rock. The

pyroclastic hanging wall ash rock was deposited, probably accompanied by the intrusion of the minor ore-cutting dikes, and followed by the deposition of the upper sediments all of which have similar prefolding remanence directions. The proposal of Jolliffe (1955) that the primary limonite or iron formation was tilted and then altered by hydrothermal solutions to goethite before folding and faulting, is considered invalid because the remanence directions of the altered ash rocks are similar to those of the unaltered ash rock rather than to those of the goethite ore. If the solutions that oxidized the limonite to goethite had also oxidized or altered the ash rock at its lowermost contact as Jolliffe suggested, then the goethite and altered ash rock should have the same remanence direction. It seems likely that the lowermost pyroclastics of the hanging wall ash rock were oxidized at the time of deposition to produce their c.r.m. immediately before deposition of the upper portion of the unoxidized pyroclastics with their t.r.m. The dikes cutting the ore were also oxidized immediately after the time of intrusion, so that their c.r.m. direction aligns parallel to the ash rock. The upper sediments presumably have a primary d.r.m.

A significant time interval of the order of hundreds of millions of years is indicated by the angular difference of about 100° of arc between the mean remanence direction of the hanging wall ash rock (plus the upper sediments) and the basic extrusives.

The basic extrusives were then deposited, probably with a prefolding t.r.m. Regional deformation caused folding, faulting, brecciation, and foliation, and intrusive activity produced veins and secondary mineralization. However, the preservation of the prefolding remanence of the goethite ore, ash rock, and other units shows that these dynamic metamorphic, thermal metamorphic, and hydrothermal effects were unimportant in the genesis of this hard goethite deposit.

CONCLUSIONS

The hard hematite ore deposit of the Cliff-Shaft mine and the hard goethite ore deposit of the Steep Rock Range were formed as laterites from the alteration of primary iron formation of the magnetite-banded oxide facies and the pyrite facies, respectively. It is proposed that the primary iron formation was exposed to the weathering action of meteoric solutions under intensely oxidizing conditions. Subsequently, the laterites were covered by the overlying formations of their respective stratigraphic sequences. Both sequences were then subjected to dynamic metamorphism in the form of folding and faulting, to thermal metamorphism in the form of intrusive activity and regional burial, and possibly to attack by hydrothermal solutions; however, the mineralogy, texture, and structure of the ores were essentially unchanged by these processes.

The south, center, and north limbs of the Caland mine's ore body have been rotated structurally relative to each other, and the Steep Rock limb or mine has been translated structurally relative to the north limb.

The postfolding remanence direction of the Keweenawan diorite of the

Cliff-Shaft mine provides a fairly reliable paleomagnetic pole position, but the remaining pole positions shown in Table II may not be reliable because tectonic rotation may have occurred for which a correction cannot be made.

More detailed conclusions about the sequence of geologic events and the paleomagnetic characteristics of the formations are given in the Discussion of Results for each of the two deposits.

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